

TOUSIF RAHMAN ANTO

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Summary

Computer Science and Engineering graduate with hands on experience building full stack web applications and AI enabled products from database schema and REST API design through to deployed UI. Comfortable owning features end to end with a background in real time systems, role based access control and secure authentication.

Education

BRAC University: Bachelor of Science in Computer Science & Engineering

*Graduated
Dhaka, Bangladesh*

- **Focus areas:** Machine Learning, Web Development, Cyber Security.
 - **Relevant coursework:** Data Structures & Algorithms, Object Oriented Programming, Databases, Operating Systems, Software Engineering, Web Development.
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Technical Skills

- **Languages:** Python, TypeScript, JavaScript, Java, C++, Dart, Lua.
 - **Frontend:** React.js, Next.js, TypeScript, Tailwind CSS, Flutter.
 - **Backend & APIs:** Node.js, Express.js, Django, Django REST Framework, REST APIs, Socket.IO.
 - **Databases & Cloud:** MongoDB, PostgreSQL, Supabase, Firebase, Vercel, Railway.
 - **AI & Engineering:** OpenAI API, Groq API, Gemini API, LangChain, prompt engineering, NLP, JWT, RBAC, CSP, Git, Docker, Postman.
 - **AI Dev Tools:** Claude, GitHub Copilot, Codex.
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Projects

MediPay: Idempotent Healthcare Billing & Settlement Engine

*GitHub
React, Node.js, Express.js, MongoDB, Decimal128, SSLCOMMERZ, Bangla QR*

- Delivered an end-to-end hospital billing workflow across 16 screens and three release milestones by translating a six-page relational data model into production-ready patient, encounter and invoice modules.
 - Prevented billing drift across partial payments and refunds by storing immutable invoice snapshots and using MongoDB **Decimal128** for exact financial calculations.
 - Reduced payment-processing risk across two checkout methods by integrating SSLCOMMERZ and Bangla QR with transactional, idempotent settlement and reconciliation.
 - Improved financial traceability across privileged workflows by generating PDF invoices and receipts, exporting CSV reports and recording role-aware audit events.
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CodeSwitch: Fault-Tolerant Multi-LLM Code Migration Engine

*Live / GitHub
React, TypeScript, Django REST, Groq, OpenAI, Gemini, Monaco Editor*

- Enabled bidirectional conversion across five programming languages by combining LLM-based transformation through Groq, OpenAI and Gemini with a deterministic regex fallback engine.
- Preserved conversion availability across three AI providers during rate limits and outages by implementing automatic provider failover and API-key rotation.
- Reduced authentication and abuse risk by building a Django REST API with httpOnly JWT cookies, brute-force lockouts and Content Security Policy middleware.
- Improved initial frontend delivery by applying route-level code splitting and lazy loading to a React 18, TypeScript and Monaco Editor interface.

The Archive: Token-Isolated Cloud Document Vault

Live / [GitHub](#)

Next.js, TypeScript, Supabase Auth, Supabase Storage, Vercel

- Delivered the document-management product end to end by designing its Supabase schema and authentication flow, building the Next.js interface and deploying it to Vercel.
- Prevented cross-user document exposure at the storage layer by enforcing authenticated uploads and per-user bearer-token isolation.
- Removed an unnecessary download step from the document-review workflow by adding secure, in-browser PDF previews.

EduCore: Headless Learning & Auto-Grading Engine

[GitHub](#)

Next.js, TypeScript, Strapi, REST APIs, RBAC

- Secured the complete learning workflow for four user types by enforcing separate Admin, Content Manager, Instructor and Student permissions in the Strapi backend.
- Unified course delivery and assessment in one platform by implementing course and lesson management, enrollment, persistent progress tracking and auto-graded quizzes.
- Reduced administrative overhead across content and user operations by building role management, platform statistics and a draft-to-publish blog workflow.

AFK Arena: Real-Time Esports Tournament Orchestration System

[GitHub](#)

React, Node.js, Express.js, MongoDB, Socket.IO, JWT

- Centralized tournament operations for a 25+ member production team by leading the end-to-end development of an esports management platform.
- Secured workflows for four stakeholder groups by building JWT-authenticated REST APIs and role-based access control for administrators, players, team managers and sponsors.
- Supported three tournament formats—single elimination, double elimination and round robin—by implementing automated bracket generation, Socket.IO notifications and live interface updates.

Research Project

Project Lead — Local Emotional-Support Conversational AI

2025 – 2026

QLoRA, Mistral, Qwen, DeepSeek, Llama, Python

- Led a five-member team in designing, developing and evaluating a locally hosted AI chatbot for empathetic, non-clinical emotional support.
- Adapted four language-model families—Mistral, Qwen, DeepSeek and Llama—with QLoRA for efficient training and deployment on consumer-grade hardware.
- Developed a response-control framework that evaluates empathy, relevance and support quality, regenerates unsuitable responses and supplies safe fallback replies.
- Integrated conversational memory, user-feedback mechanisms and privacy safeguards, including identifier redaction and pseudonymous session tracking.
- Evaluated performance using 37 matched prompt pairs and a separate 360-turn conversational study; compared automated metrics with three AI-judge collections across 320 shared responses.
- Identified limitations in repeated responses, fallback behavior and evaluation consistency, documenting the findings in an IEEE-format manuscript with reproducible analysis code.

Professional Experience

Dreamers Academy: Coding Instructor & Curriculum Developer

2024 – 2026

Dhaka, Bangladesh

- Taught beginner Python programming including syntax, control flow, functions and structured problem solving.
- Created a Lua curriculum for Roblox Studio game development and provided individual learner feedback.

Leadership & Awards

Millennium Fellowship Program (UNAI & MCN)

2025 – 2026

- **Fellow, Class of 2025:** Selected from 60,396 worldwide applicants; now serving as a **GAC Member** evaluating global applications via a data driven, rubric based scoring system. Certificate of Completion
- GAC Certificate of Appreciation [[click to view](#)]
- Letter of Recommendation (MCN Admissions Manager)